

Dristanta Das

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SKILLS

AI/ML: NLP, LLMs, NER, RAG, Machine Learning, Deep Learning, Statistical & Predictive Modeling

Frameworks: LangChain, LangGraph, PyTorch, Hugging Face, Scikit-learn, XGBoost, FastAPI, Agentic AI

Tools & Cloud: Python, R, SQL, Git, Docker, BigQuery, Vector Databases, AWS, Azure

Design: Architecture Design, Data Flow Modeling, UML, Anomaly Detection, HIPAA/GDPR Compliance

EXPERIENCE

Genpact

Bengaluru, Karnataka

Senior Data Scientist

Apr 2025 – Present

Conversational Data Insights

- Architected and deployed an end-to-end **Text-to-SQL** system using LangChain and LangGraph that converts natural-language questions into SQL and returns concise natural-language summaries of the results, supporting 50+ beta users with **10** concurrent sessions and cutting query-formulation time by **65%**.
- Engineered robust agentic workflows with state management, error handling, and dynamic routing (**92%** task-completion rate), then drove their integration and scalability into production with **12+** cross-functional teams, sustaining **500+** weekly queries while improving validation, reliability, and client satisfaction.

Invoice Processing

- Spearheaded an automated invoice-processing pipeline using LangChain and Azure OpenAI, achieving **85%+** extraction accuracy across 10,000+ invoices and reducing manual processing time by **60%**.
- Extracted key fields from diverse invoice formats across EMEA and NAM regions with support for **8+** languages, improving accuracy and consistency in information retrieval.

UST

Kolkata, West Bengal

Associate III Data Scientist

Oct 2022 – Mar 2025

- Orchestrated development of a client-facing provider search system using advanced NLP techniques and open-source LLMs, boosting search efficiency by **30%** and accelerating data preparation tasks by **40%**.
- Implemented NER and semantic search capabilities for retrieving medical terms from layperson language, achieving a **25%** improvement in operational efficiency and improving service quality for end-users.

Associate II Data Scientist

Jul 2022 – Sept 2022

- Conducted a patient data evaluation using **BigQuery** with comprehensive data transformations and statistical analysis, improving the client's understanding of adherence drivers by **45%** and improving outreach to more than 10,000 patients.
- Designed and implemented a data anomaly detection system for categorical healthcare data, adhering to HIPAA and GDPR compliance standards.

PERSONAL PROJECTS

Retrieval-Augmented Generation (RAG) QA Bot Development

- Engineered a Python-based QA bot with hybrid retrieval using **BM25** and **DPR**, achieving **40%** higher answer accuracy compared to traditional search methods.
- Integrated **GPT-3.5-turbo**, **Phi-2**, and **Llama3** with contextual compression to reduce hallucinations by 60%; optimized performance using SQLite caching and Slack integration.

Automated Summary Evaluation Competition

- Organized by *CommonLit*, *Vanderbilt University*, *Georgia State University* & *Kaggle*
- Developed a machine learning model that gained a **0.48** Mean Columnar Root Mean Squared Error, ranking in the top **9%** of participants.
- Achieved a Bronze Medal on Kaggle by developing a solution for evaluating student-written summaries, enhancing evaluation accuracy by **65%** and outperforming **90%** of competing solutions.

EDUCATION

RKMVERI, Belur

July 2020 – Jun 2022

M.Sc. in Big Data Analytics

CGPA: 7.71/10.0

Presidency University, Kolkata

May 2017 – Jun 2020

B.Sc. in Mathematics

CGPA: 7.05/10.0

CERTIFICATE & ACHIEVEMENTS

AWS Certified Cloud Practitioner: CLF C01

Microsoft Certified: Azure Fundamentals, AZ-900